Student Name:Saparbay Amir Student ID:230103037

Session / Date:09.10.2026 Assigned GPU ID: 0

- **NVIDIA Driver Version:** `580.82.07`
- **CUDA Version Supported by Driver:** `13.0`
- **Allocated GPU Name:** `Tesla T4`
- **Total Available VRAM:** `15360 MiB`
- **Compute Capability Major.Minor:** `7.5`
- **Max Threads Per Block:** `1024`

Task 2

```
N           | CPU (ms)    | GPU Total (ms)   | GPU Kernel Only (ms) | Winner
-------------------------------------------------------------------------
100         | 0.046       | 0.855            | 0.316                | CPU
1000        | 0.236       | 0.468            | 0.059                | CPU
10000       | 2.358       | 0.428            | 0.058                | GPU
100000      | 25.405      | 0.842            | 0.097                | GPU
1000000     | 356.554     | 5.354            | 0.133                | GPU
5000000     | nan         | 14.816           | 0.293                | GPU
10000000    | nan         | 62.491           | 0.518                | GPU
```

Task 3

```
Launching defective kernel with 1024 threads on 1000-element array...
Execution finished (or did it crash?). Checking host copy...
Elements processed: 1000. Last element: 999.0
```

```
Global ID  | Block ID   | Thread ID  | Status
-------------------------------------------------
32         | 4          | 0          | ACTIVE
33         | 4          | 1          | ACTIVE
34         | 4          | 2          | ACTIVE
35         | 4          | 3          | ACTIVE
36         | 4          | 4          | ACTIVE
37         | 4          | 5          | IDLE (GUARDED)
38         | 4          | 6          | IDLE (GUARDED)
39         | 4          | 7          | IDLE (GUARDED)
```

Task 4

```
--- RUNNING NAIVE KERNEL (5 TRIALS) ---
Trial 1 | Target: 50000.0 | Naive Kernel Output: 2.0 | Deviation: 49998.0
Trial 2 | Target: 50000.0 | Naive Kernel Output: 2.0 | Deviation: 49998.0
Trial 3 | Target: 50000.0 | Naive Kernel Output: 2.0 | Deviation: 49998.0
Trial 4 | Target: 50000.0 | Naive Kernel Output: 2.0 | Deviation: 49998.0
Trial 5 | Target: 50000.0 | Naive Kernel Output: 2.0 | Deviation: 49998.0

--- ATOMIC KERNEL RESULTS ---
Atomic Sum Output: 50000.0
Naive Time: 0.460 ms vs Atomic Time: 185.305 ms
```

Task 5

- **Grid Dimensions:**
  - **Blocks in X:** 64 $\left(\lceil 1024 / 16 \rceil\right)$
  - **Blocks in Y:** 64 $\left(\lceil 1024 / 16 \rceil\right)$
  - **Total Blocks:** 4096 (64×64)
- **Verified Pixel Value at Center Point (512, 512):** 1.000
- **Verified Pixel Value at Outer Corner Point (0, 0):** 0.000

Task 6

```
============================================================
OFFICIAL VERIFICATION CHECKSUM TOKEN: 252B66AA8B4DCBAC
============================================================
252B66AA8B4DCBAC
```